

BRIDGE INSPECTION REPORT

Bridge No. 12345 — Element-Level Condition Assessment

Inspection Date: 2026-04-15

This report is organized by element. Each element table lists the condition state quantities (CS1=Good, CS2=Fair, CS3=Poor, CS4=Severe) for each location/span along with inspector notes.

Element 12: Reinforced Concrete Deck

Unit: sq.ft. Total Quantity: 24500

Location	CS1	CS2	CS3	CS4	Notes
Span 1	4800	100	0	0	Minor map cracking on underside near west curb.
Span 2	4600	280	20	0	Efflorescence visible. Three spalling areas ~6" at midspan.
Span 3	4500	380	20	0	Heavy efflorescence. Transverse cracks every 8-10 ft.
Sp. 4	4700	200	0	0	Light scaling on top surface near drains.
Span 5	4900	100	0	0	Good condition overall.

Element 107: Steel Open Girder/Beam

Unit: lin.ft. Total Quantity: 4200

Location	CS1	CS2	CS3	CS4	Notes
Span 1	820	30	0	0	Paint intact. Minor surface rust at bearings.
Span 2	810	80	10	0	Section loss G3 bottom flange near Pier 2 — 1/16" deep over 18".
Span 3	790	130	30	0	Pack rust between top flange and deck haunch. Paint peeling G2-G4.
Span 4	820	40	0	0	Minor paint loss at splice plates.
Span 5	820	20	0	0	Good condition.

Element 215: Reinforced Concrete Abutment

Unit: lin.ft. Total Quantity: 120

Location	CS1	CS2	CS3	CS4	Notes
Abutment 1 (West)	50	8	2	0	Vertical crack at NW corner, ~1/16" wide, 4 ft tall.
Abutment 2 (East)	55	5	0	0	Light staining. No structural concerns.

Element 205: Reinforced Concrete Column

Unit: each Total Quantity: 16

Location	CS1	CS2	CS3	CS4	Notes
Pier 1 (Bent 2)	4	0	0	0	Good condition all columns.
Pier 2 (Bent 3)	3	1	0	0	Column C2 has spall at base ~8"x6", rebar exposed.
Pier 3 (Bent 4)	4	0	0	0	Good condition.
Pier 4 (Bent 5)	3	1	0	0	Column C4 minor honeycomb at construction joint.

Element 234: Reinforced Concrete Pier Cap

Unit: lin.ft. Total Quantity: 240

Location	CS1	CS2	CS3	CS4	Notes
Bent 2	58	2	0	0	Minor shrinkage cracks.
Bent 3	50	8	2	0	Diagonal shear crack near south end, ~1/8" wide. MONITOR.
Bent 4	56	4	0	0	Light efflorescence below bearing seats.
Bent 5	55	5	0	0	Surface scaling on east face.

Element 310: Elastomeric Bearing

Unit: each Total Quantity: 24

Location	CS1	CS2	CS3	CS4	Notes
Abutment 1	4	0	0	0	Bearings seated correctly. No tearing.
Pier 1	3	1	0	0	Bearing B2-N3 shows slight bulge. Monitor.
Pier 2	4	0	0	0	Good condition.
Pier 3	3	1	0	0	Bearing B4-S1 has minor edge cracking on elastomer.
Pier 4	4	0	0	0	Good condition.
Abutment 2	4	0	0	0	Bearings clean and properly aligned.

Element 300: Strip Seal Expansion Joint

Unit: lin.ft. Total Quantity: 180

Location	CS1	CS2	CS3	CS4	Notes
Abutment 1	30	0	0	0	Clean, intact.
Pier 1	28	2	0	0	Minor debris accumulation.
Pier 2	25	5	0	0	Seal lifted in 6" section. Recommend cleaning.
Pier 3	28	2	0	0	OK.
Pier 4	30	0	0	0	Clean.
Abutment 2	30	0	0	0	Clean.

Element 330: Metal Bridge Railing

Unit: lin.ft. Total Quantity: 1800

Location	CS1	CS2	CS3	CS4	Notes
Span 1	350	10	0	0	Paint intact.
Span 2	340	18	2	0	Impact damage to top rail at midspan, ~3 ft section dented.
Span 3	345	15	0	0	Minor paint loss at posts.
Span 4	348	12	0	0	Good condition.
Span 5	350	10	0	0	Good condition.

Element 510: Wearing Surfaces

Unit: sq.ft. Total Quantity: 24500

Location	CS1	CS2	CS3	CS4	Notes
Span 1	4700	200	0	0	Light surface scaling.
Span 2	4400	480	20	0	Pothole patches visible. 2x rutting in WB lane.
Span 3	4500	400	0	0	Surface scaling.
Span 4	4700	200	0	0	Good condition.
Span 5	4900	100	0	0	Good condition.

Element 8001: Steel Diaphragm

Unit: each Total Quantity: 40

Location	CS1	CS2	CS3	CS4	Notes
Span 1	8	0	0	0	All connections sound.
Span 2	7	1	0	0	One bolt missing at D3.
Span 3	8	0	0	0	Good.
Span 4	7	1	0	0	Loose bolt at D7.
Span 5	8	0	0	0	Good.

Element OW-44: Cathodic Protection System

Unit: L.S. Total Quantity: 1

Location	CS1	CS2	CS3	CS4	Notes
Whole Bridge	0	1	0	0	System operational but anode reading low. Schedule replacement.